

Ariel Levinson

Aventura, FL 33180 | (305) 318-7372 | levinson.a@northeastern.edu | <https://www.linkedin.com/in/ariel-levinson>

SUMMARY OF QUALIFICATIONS

Bioinformatics Master's graduate from Northeastern University with experience in clinical research, data standards, and analysis. Proficient in R and Python programming and Bash scripting for computational analysis in biological research. Experienced in presenting scientific and statistical concepts in academic settings.

EDUCATION

Northeastern University, Boston, MA

December 2025

Master of Science in Bioinformatics

GPA: 4.0

Coursework: Genomics in Bioinformatics, Bioinformatics Methods 1 & 2, Bioinformatics Programming, Statistics for Bioinformatics, Omics in Bioinformatics, Collecting, Storing, and Retrieving Data

Florida International University, Miami, Florida

December 2019

Bachelor of Science in Biological Sciences & Bachelor of Arts in Natural and Applied Sciences

GPA: 3.92

Coursework: UG Bioinformatics for Biologists, Genetics, Cell Biology

SKILLS

Programming Languages: Python, R, Bash

Bioinformatics Tools: Nextflow, BLAST, GSNAP, GMAP, BioPerl, Trimmomatic, SAMtools, TransDecoder, SPAdes assembler, QUAST, BWA, ClustalW, ClustalO, Bowtie2, DESeq2, FastQC, MultiQC, edgeR, GATK, SQLite

Bioinformatics Skills: Programming predictive models for genetic datasets, building multiple sequence alignments, parsing FASTA and FASTQ files, managing version control using Git, working in HPC environments using workload managers (e.g., Slurm), statistical analysis of RNA-Seq and other NGS datasets.

Databases: UniProt/SwissProt, NCBI, KEGG, Gene Ontology (GO), Ensembl, GEO, PDB

Laboratory Techniques: Gel electrophoresis, PCR, DNA and RNA purification, western blotting, rearing *Drosophila* fly model organisms, dosage preparation, RNA-Seq

Operating Systems: Linux, Windows, Mac

Languages: English, Spanish, Portuguese.

ACADEMIC PROJECTS

Studying LCK Through Multi-Omics Approaches, Northeastern University, Boston, MA

Jan 2025 – Apr 2025

- Proposed an NIH-style research plan to investigate how distinct Lck proteoforms differentially influence T cell activation and disease progression using multi-omics approaches (epigenomics, proteomics, metabolomics, lipidomics).
- Created simulated data visualizations, outlined analytical pipelines, and assessed feasibility through cost and sustainability analysis to guide project design.

WORK EXPERIENCE

Science & Statistics Tutor, Florida International University, Miami, FL

Aug 2020 – Jan 2022

- Designed and presented statistics-focused workshops to supplement coursework, improving student comprehension of key methods (z-tests, t-tests, chi-square, simple and multiple linear regression).
- Facilitated the learning of FIU-enrolled students in the subjects of biology, chemistry, and statistics through one-on-one and group tutoring sessions to promote higher average student performance and educational attainment.

RESEARCH EXPERIENCE

Undergraduate Volunteer Research Assistant, Florida International University, Miami, FL

Feb 2018 – Dec 2018

- Collaborated with PI to design experiments on the role of nutrition in the development of Parkinson's disease symptoms using *Drosophila* models, leading to a refined experimental approach.
- Collected and analyzed six months of data using chi-square tests to assess treatment effects on disease progression.
- Instructed other students on effectively performing the experiment, ensuring the long-term viability of the studies.